

# How a school swarms

collective motion for the aquarium level's ten followers · the Couzin zone model, measured against our own Forth implementation

Lengths are in **body lengths** (BL; a fish is 3.5 in). The tank is  $13.4 \times 3.4 \times 4.7$  BL inside. A **school** is polarised (p\_group near 1); a **swarm** is not (p near 0.1): moving one zone width,  $\Delta r_o$ , switches between them.

## A · Four collective states from one rule, simulated here (N = 100, the paper's $r_r, \alpha, \theta, s, \sigma$ )

A · swarm



top view

B · torus



spin-axis view

C · dynamic parallel



side view

D · highly parallel



side view

$\Delta r_o = 0, \Delta r_a = 15$   
p\_group 0.16 · m\_group 0.10

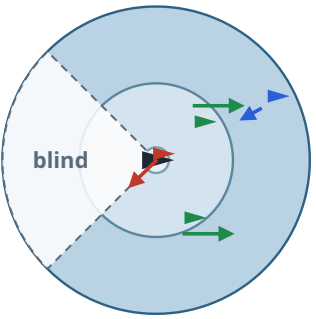
$\Delta r_o = 1, \Delta r_a = 12$   
p\_group 0.09 · m\_group 0.90

$\Delta r_o = 3, \Delta r_a = 10$   
p\_group 0.90 · m\_group 0.03

$\Delta r_o = 10, \Delta r_a = 10$   
p\_group 1.00 · m\_group 0.02

ours · N = 100 simulated here; the zone widths are ours, the paper prints none

## B · The model: three zones and a blind volume



repulsion zone,  $r_r = 1$   
anyone here: turn away, nothing else counts

orientation zone, width  $\Delta r_o$   
those here: match their heading

attraction zone, width  $\Delta r_a$   
those here: turn toward them

blind volume, 90° behind  
 $\alpha = 270^\circ$ : cannot see here

Drawn at the school setting:  $\Delta r_o = 5, \Delta r_a = 6$

verified · zones,  $\alpha, r_r$ : Couzin 2002

ours · widths drawn

## C · The map: p\_group over the two zone widths

attraction width  $\Delta r_a$

orientation width  $\Delta r_o$  (BL)

0 1 2 3 4 6 8 10 12 15

|    |   |   |   |   |   |   |   |   |   |   |
|----|---|---|---|---|---|---|---|---|---|---|
| 15 | S | T | ? | D | D | P | P | P | P | P |
| 12 | S | T | ? | D | D | P | P | P | P | P |
| 10 | S | ? | D | D | D | P | P | P | P | P |
| 8  | S | F | F | D | D | P | P | P | P | P |
| 6  | F | F | F | D | D | P | P | P | P | P |
| 4  | F | F | F | D | D | P | P | P | P | P |
| 3  | F | F | F | F | D | P | P | P | P | P |
| 2  | F | F | F | F | D | P | P | P | P | P |
| 1  | F | F | F | F | F | P | P | P | P | P |
| 0  | F | F | F | F | F | F | P | P | P | P |

p\_group (alignment)

< 0.25 < 0.6 < 0.9 ≥ 0.9

S swarm: low p, low m

T torus: low p, high m

D dynamic parallel: high p

P highly parallel: p ≈ 1

F fragments (more than half)

? seeds disagree: no one state

our swarm, torus, school

ours · N = 100, 3 replicates, 700 steps (paper: 30)

## D · The rule in eight lines, and the Forth word that does each

1 See the others, except behind

2 Repulsion if anyone is within  $r_r$

3 Else orientation and attraction

4 Nobody in any zone: keep the heading

5 Turn toward d by at most  $\theta\tau$ , move s  $\tau$

6 The leader counts w times in sums 3

7 Walls repel like a neighbour

8 A dart startles the followers nearby

$u_{ij} \cdot v_i \geq \cos(\alpha/2)$

$d = -\sum u_{ij}$  (nothing else counts)

$d_o = \sum v_j, d_a = \sum u_{ij}; d = (\hat{d}_o + \hat{d}_a)/2$

$d = v_i$

$v \leftarrow \text{rotate}(v, d, \theta\tau); c \leftarrow c + s \tau v$

$\sum w_j v_j, \sum w_j u_{ij}$

wall within 0.6 BL: one more repulsion

swim away from the leader for 0.5 s

sch-pair

sch-pair

sch-pair, sch-want

sch-want

sch-turn, sch-follow

sch-pair (weight)

sch-wall

sch-startle-all

The paper's random error ( $\sigma = 0.05$  rad) is left out of the Forth: the caller adds it. Our units differ from the paper's, on purpose: at its  $s = 3$  BL/s and  $\theta = 40^\circ/s$  a  $90^\circ$  turn takes 6.75 BL of swimming (radius 4.3 BL), wider than the tank is deep (3.4 BL), and the tank runs below could not keep the fish in. Ours:  $s = 2$  BL/s,  $\theta = 120^\circ/s$ , 1.50 BL, radius 0.95 BL.

verified · rules 1–5 and the paper's s,  $\theta$ : Couzin 2002, eqns 1–3 and Fig. 3

ours · rules 6–8, our s,  $\theta$ , the 0.6 BL wall, the 0.5 s startle

## F · The Forth core, school.fth: the four rule words, and what they cost

| sch-pair                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 168 B · 24 lines                                                                                                                                                                                                                                                                                                    | sch-want                                                                                                                                                                                                                                                                                                                                                | 88 B · 11 lines                                                                                                                                                                                                                                                                                                                                                                                             | sch-turn | 104 B · 12 lines | sch-follow | 129 B · 16 lines |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------|------------|------------------|
| one neighbour: blind volume, then<br>repulsion, orientation or attraction<br>: sch-pair ( j -- )<br>MB_YOU sc!<br>MB_X MB_U you-delta<br>MB_Y MB_U 1 + you-delta<br>MB_Z MB_U 2 + you-delta<br>MB_U vlen2 dup 0.000001 ><br>over reach2 < &<br>if<br>sch-sqrt dup MB_DIST sc!<br>1 swap / MB_U vscale<br>heading-dot-u MB_COST par@ >=<br>if<br>r-rep inside if<br>-1 MB_SUM_R acc+u<br>MB_N_R count+<br>else r-ori inside if<br>you-weight MB_SUM_0 acc+v<br>MB_N_0 count+<br>else<br>you-weight MB_SUM_A acc+u<br>MB_N_A count+<br>then then<br>then<br>else drop then ; | the wanted direction from the three sums, or the<br>old heading<br>: sch-want<br>MB_WANT vzero<br>MB_N_R seen? if<br>MB_SUM_R add-wanted<br>else<br>MB_N_0 seen? if MB_SUM_0 add-wanted then<br>MB_N_A seen? if MB_SUM_A add-wanted then<br>then<br>MB_WANT vnorm 0.000001 > not if<br>MB_WANT heading>sc<br>then ; | turn toward it by at most $\theta\tau$ , keep the<br>heading unit length<br>: sch-turn<br>MB_VX me sch@ MB_WANT sc@ *<br>MB_VY me sch@ MB_WANT 1 + sc@ * +<br>MB_VZ me sch@ MB_WANT 2 + sc@ * +<br>dup MB_COST par@ >= if drop else<br>perp-all<br>MB_PERP vnorm 0.000001 > not if<br>perp-fallback<br>then<br>turn-all<br>MB_WANT vnorm drop<br>then ; | one follower, start to finish: scan,<br>walls, startle, turn, move<br>: sch-follow<br>MB_ME sc!<br>clear-sums<br>sch-n 0 do<br>i me <> if i sch-pair then<br>loop<br>sch-walls sch-want<br>sch-startle? if<br>sch-startle-away<br>sch-startle-tick<br>then<br>sch-turn<br>MB_WANT sc@ MB_NVX me sch!<br>MB_WANT 1 + sc@ MB_NVY me sch!<br>MB_WANT 2 + sc@ MB_NVZ me sch!<br>0 advance 1 advance 2 advance ; |          |                  |            |                  |

2936 B

of the 64 KB dictionary (4.5 %): all 70 words, 193 code lines

7.0 ms

one tick, 11 fish, on the Chromecast HD (median of 3 runs); x86 here 9.3 ms

1e-03

largest heading error against couzin.py in one tick (the engine's sine is 0.2 % off)

~0.7 ms

per follower per tick: spread over frames, the full 10 Hz step is about 7 % of a core

ours · the Forth, its size, error and timing are our measurements: the interpreter only, not yet inside the engine

## G · The Forth, running in the tank: swarm, torus, school, and a startle

swarm setting (0, 10)



leader weight w

dist · align · p · in box

w = 1 2.7 BL · +0.02 · 0.27 · yes

w = 3 2.8 BL · -0.12 · 0.40 · no

w = 6 2.5 BL · +0.07 · 0.51 · no

torus setting (1, 10)



leader weight w

dist · align · p · in box

w = 1 3.4 BL · -0.05 · 0.58 · yes

w = 3 2.3 BL · +0.24 · 0.64 · no

w = 6 2.3 BL · +0.31 · 0.61 · no

school setting (5, 6)



leader weight w

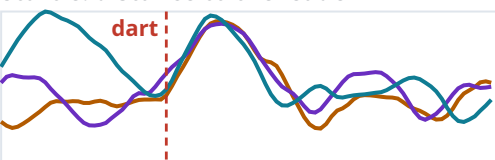
dist · align · p · in box

w = 1 3.8 BL · +0.12 · 0.87 · yes

w = 3 3.5 BL · +0.17 · 0.84 · yes

w = 6 3.2 BL · +0.18 · 0.82 · yes

Startle: distance to the leader



swarm torus school

–6 s to +12 s; top = 6 BL; one seed

ours · the Forth core in the tank's real box (w = 3, seed 1 drawn; the table is the mean of 3 seeds; "no" = a fish strayed more than 0.5 BL out)

## Sources · live links in the PDF; what could not be verified is marked

S1 Couzin et al. 2002, J. Theor. Biol.

verified

zone model, states, metrics, Fig. 3

S2 Couzin et al. 2005, Nature

unverified

informed minority guides a group

S3 Pitcher 1983, via Wikipedia

unverified

shoal to school: a continuum

S4 Ocellaris clownfish (Florida Museum)

unverified

site-attached: not schooling fish

S5 the aquarium level and this repository

ours

tank, fish, zForth, our runs

How to read the chips

verified

the source was opened; the number is on its page

unverified

from a summary, or not opened

ours

our own choice, maths or measurement

Not claimed: that real clownfish school (they do not), that our sweep shows the paper's hysteresis (it is too coarse), or that the Forth was timed inside the engine. Data dated 2026-10-01.